

Monetary sanctions and deterrence: the effect of speeding tickets on road behavior

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Abstract: Governments use monetary fines in order to deter socially unwanted behavior on a wide array of issues. It is commonly assumed that, fixing the probability of sanction, the higher the price, the more deterrent a sanction is. I use data from automated speed cameras in Bogotá, Colombia to measure the deterrent effect of a speeding ticket. By using the real speed limit as a source of exogenous variation and a regression discontinuity design, I find that speeding tickets have a general deterrent effect on speeding behavior but a null effect on reducing traffic accidents. I show that the deterrent effect is concentrated on cars, and absent on motorcycles. Using the commercial price of the vehicle as a proxy for wealth and the fact that the monetary sanction is fixed, I find that the deterrent effect is concentrated on quintile four of the distribution. I validate these findings with a judge-IV using the fact that all tickets are manually verified by a police agent. In short, tickets might be less useful for preventing road fatalities than commonly assumed and inefficiently expensive for a significant number of drivers.

1. Introduction:

A central question for governments is how to improve the deterrent power of their legal sanctions. A common avenue is to increase the monetary sanctions under the assumption that a higher expected cost results in a lesser prevalence of unwanted behavior. In context with little enforcement this assumption might be wrong; legal sanctions can be too expensive for certain actors that not only avoid paying the fine but also continue with unwanted behavior.

This paper explores the question if a legal monetary sanction can be too expensive. I use traffic tickets and accidents to answer it. I explore the differential effect of relative price on behavior in a context in which there is a fixed price for a speeding infraction, but it can be compared to the market price of the vehicle. The possible selection of drivers into law breakers and law-abiding drivers is dealt with

by using a regression discontinuity design with the unknown gap between the posted street limit and the real speed at which speeding tickets are given.

This article starts by answering questions on the general causal effect of speeding tickets on behavior: What is the causal effect of a speeding ticket on subsequent behavior of a driver? The most studied subsequent behavior usually speeding itself, but with the data available I can also observe the chances of being involved in an accident. Since speeding tickets are used as a tool for preventing road accidents, this is the most relevant variable to study. I will argue that studying the effect of traffic tickets on subsequent speed tickets is like measuring the effect of tacking a pill to reduce cholesterol on taking the next pill, not on cholesterol levels.

This paper relates to the literature on financial fines and fees (commonly referred to as legal financial obligations, or LFOs). There is debate on the capacity of higher fines to increase revenue and the effect this has on recidivism. Some papers find a modest effect on revenue at the cost of higher recidivism (Giles, 2023). Others argue that even if the effect of sanctions is higher revenue, they have no effect on recidivism and thus are not justifiable ways of increasing deterrence (Finlay et al., 2024). The authors argue that the null effect in all relevant criminal behaviors is precisely estimated, not to lack of power. In recent work, (Norris & Rose, 2024) argue that reducing LFOs will impact municipal revenues without increasing deterrence. The evidence provided by (Bing et al., 2024) on a randomized payment of existing fines for criminals shows that this policy only reduces recidivism related to the financial costs of the LFO itself. These effects are not negligible and point to no deterrent effect at a significant cost of engagement with the criminal system. In summary, LFO's seem to have a null effect on criminal behavior, while a modest effect on increasing municipal revenue. Questions of fairness arise from these two facts.

This question has also been examined in the specific case of traffic tickets, but mainly through the lens of revenue. Some have shown that increasing fines has only a modest effect on timely payment (Traxler & Dušek, 2023) and recidivism (Dušek & Traxler, 2022). A very detailed of the effect of the price of ticket fines is (Kaila, 2024), who uses the relative to income price of traffic tickets in Finland to shows that there is in fact a reduction on recidivism for higher fines, but the effect is short lived and the fines need to be significantly more expensive. In their work, (Goncalves & Mello, 2024) also study the effect of harsher sanctions by exploiting the discretionary sanctions given by police officers. They find that harsher fines do reduce recidivism and, to a lesser degree, crash involvement. They also find that officers discretion is inefficient since harsher sanctions are given to drivers that are less deterred by them. Contributing to this new evidence on the indirect deterrent effect of traffic fines on road safety is one of the central contributions of this paper.

Another branch of the literatures studies what citizens learn by interacting with law enforcement. Using a change on the punishment for recidivism, (Philippe, 2024)

finds that learning about criminal law appears to be limited to people with direct involvement with the law. In the same direction, it has been shown that individuals do adjust their priors after receiving a speeding ticket (Dušek & Traxler, 2022). As I will explain later, I contribute to this literature by studying subsequent behavior where there is going to be future interactions with the law and also where there is not. I can speak to how interactions with the law affects behavior when citizens know the law is not present.

Finally, this paper also contributes to a broader literature on the effect of traffic enforcement on behaviors (De Angelo & Hansen, 2014; Hansen, 2015; Luca, 2015; Makowsky & Stratmann, 2011). In particular, my works add to a growing literature that studies fines and enforcement in relation to factors that do not affect either sanctions or probability of enforcement but have been shown to have an effect on traffic accidents. Things like the political cycle (Bertoli & Grembi, 2021), or the stock market (Fry & Farrell, 2023) (Giulietti et al., 2020). Close to this strand is a set of papers that explores behavioral responses to traffic enforcement and tries to derive policy recommendations to reduce road accidents and fatalities (Zhang et al., 2020) (Lu et al., 2016). My work contributes by modelling and measuring perceptions of fairness of sanctions as a mediator of changes in enforcement. I also contribute to this literature by adding evidence from a city in the developing world, something that is still rare in the field.

Examining the effect of speeding tickets on behavior is difficult because of selection. Drivers that decide to exceed speed limits might be significantly different in unobservable characteristics that also relate to risky behavior, thus comparison between ticketed and not ticketed drivers cannot capture the causal effect of tickets. But the real working of the process for giving speed tickets offers a way to solve this issue. Speed tickets are not given to all vehicles that exceed the maximum speed posted on road signs, but to a different speed with an additional tolerance. Since the number of kilometers per hour of this tolerance is not known to drivers, thus a discontinuity exists. A regression discontinuity design will be used to make a valid comparison between only drivers that decide to exceed the speed limit, but some receive a ticket and some do not. Evidence for the validity of the design will be provided on the methodology chapter of this paper.

The paper proceeds as follows. Section 2 presents the institutional background of speed fines in Bogotá, Colombia and the Data that will be used on the estimation. Section 3 presents a model of individual behavior that presents the hypothesis to be tested. Section 4 discusses the estimation strategy and presents the evidence necessary for its validity. Section 5 presents all the results from the estimation. Section 6 presents some robustness checks on the main estimation. Section 7 discusses the main findings in relation to the behavioral model and derives possible policy recommendations.

2. Institutional background and data:

Automated enforcement of speed started in Bogotá in 2020, with the staggered installation of Cameras in the most dangerous zones. The program was originally called *Cámaras Salvavidas* (Spanish for Lifesaving cameras) to emphasize that the objective of the program was not municipal revenue but behavioral change. After some debate on the legality of the fines, Colombia's higher court¹ allowed for the fine to be given to the owner of the vehicle. Two elements of the legal frame are crucial in the rest of the paper. First, tickets have a fixed price independent² of the type of vehicle or the income of the owner, or the speed at which a vehicle is circulating. Second, there is no additional fine for recidivism.

The central database I will use for defining treatment includes all vehicles that exceeded the legal speed limit under a Speed Camera from August 16 to October 15, 2024. This dataset is comprised of 181,489 distinct vehicles that passed the speed Cameras at the time at which they give speeding tickets, which corresponds to daylight. Since I will only be using vehicles that passed under the Cameras once a total of 158,703 comprise the full sample used for estimation³. This is equivalent to 87% of the plates that exceeded the limit. I will call this sample my unmatched sample. On the Appendix section 2 shows some results using also some vehicles that passed more than once.

I can also access the databases for all existing tickets and accidents occurring in Bogotá. I can see if each plate received a ticket or was involved in an accident both before and after the period of study, until March 15 of 2025. Those two will be my main outcome variables but also are important in correcting for specific characteristics of the sample.

I can access the registration record for all vehicles in Bogotá and match the plate with information on brand, model and year of the vehicle. A web scrapping procedure of the most used website for selling used cars in the city is done to obtain the commercial value of each brand -model-year combination⁴. This leaves me with a total of 62,288 vehicles for which I can find the commercial prices, naturally a subsample but is big enough for the estimation to have statistical power. I will use the price of the vehicle as a proxy for wealth. The heterogeneous effect of the relative price of the ticket can thus be estimated. Local effects and external

¹ This is ruling C-321 of 2022 by Corte Constitucional del Colombia.

² Technically there is an early payment, but all vehicles are subject to this. So the price is the same for everyone.

³ Full distribution of the number of observation by vehicle in this sample is shown on Table 1 of the Appendix. This is a particular problem since vehicles might circulate one time below the real limit and another one above the real limit, thus making it harder to classify them as treated or untreated in terms of the discontinuity. I will do robustness checks on this, but mostly just use vehicles that circulated once.

⁴ All data on vehicle prices is taken from [tucarro.com](https://www.tucarro.com), the most used website for selling used vehicles.

validity questions will be analyzed in the Results section. I will call this sample my matched sample.

In no way am I arguing that my matched sample is similar to the Full Sample. The decision to register the vehicle in Bogotá is surely related to behavior as the fact that the registry is growing in surrounding municipalities but not in Bogotá. This is particularly caused by the decision of new motorcycle owners to register their vehicle outside Bogotá, even if this is where they plan to use it. Thus, the effect seen on that sample might only extend to other drivers that decide to register their vehicles in Bogotá.

3. A model of individual behavior.

One of the key contributions of this paper is looking besides timely payment when measuring the behavioral effects of receiving a fine. I will also look at recidivism in speeding as a measure of direct deterrence and to the chance of being in an accident as indirect deterrence of a fine. All these effect should be different depending on the wealth of a driver, as my main research question suggests.

I now present a model of individual behavior in which the decision of timely payment and of recidivism are affected by wealth. The model is supported in a very clear fact of how speeding fines work in the specific context of Bogotá, that is very common on similar cities. Almost all the speeding fines in the city are given by Cameras and these Cameras are highly visible by law⁵. Thus, the perceived probability of getting a ticket if speeding where cameras are located is 1. On the contrary, and also a result of Cameras being present and visible, the perceived probability of receiving a speeding fine in the rest of the city is essentially 0. In short, Cameras have a direct deterrent effect on the spot where they are located and an indirect behavioral effect, with no deterrence, in the rest of the city.

This fact leads me to the first premise of my model:

Premise 1: After receiving a fine, the decision to speed on the spots where Cameras are located (reoffend) is independent of the decision to speed (change behavior) on the rest of the city.

This allows me to have all four possible combination of the two decisions. A fine can reduce speeding behavior on the location of the cameras, while increasing speeding behavior on the rest of the city. It could also increase speeding both in where cameras are located and where there are no cameras.

As my research question suggests, I am interested in the heterogenous effect of a ticket depending on the wealth of each driver. The market price of the vehicle is my

⁵ In May 2024 there was a city wide intervention to make the Cameras even more visible. The metal posts where the cameras are installed were painted in bright colors and new red signs were painted on the road with the speed limit. Unlike the start of the program, by 2024 Speed Cameras were designed to not surprise any driver.

main measure of wealth. In most of the application I will use the quintiles of the distribution of vehicle prices as a proxy for the distribution of wealth, but my results are sensible to using different elements of the distribution.

Reoffending:

I will call the decision to drive above the speed limit on the location of the Cameras as reoffending. Note that the model will use the fact that the real limit is different than the posted limit, but for this decision the posted limit is the one I will use. Drivers are not aware of the real limit, as I will show soon.

The benefit of not reoffending is being able to do the things that having an unpaid fine prohibits. In the case of Colombia this is associated with two things: the ability to sell the vehicle and the avoidance of a seize of one's salary by the local government⁶. In short, the benefit of not reoffending is associated with being part of the formal economy, that I will call *formality*. This leads to the second premise of the model.

Premise 2: After receiving a fine, the decision to speed on the spots where Cameras are located is the result of the comparison between the perceived individual benefits of legality compared to the relative monetary cost of the fine plus the utility of speeding.

In Colombia⁷ the amount of a fine only depends on the type of infraction. All drivers that speed must pay the same amount: F , for fine. Now the real price is relative: F divided by personal wealth. This is added to the utility of speeding that is only a function of driving at a higher speed $\bar{S}$ than the limit S . Boths speeds discount the risks associated with increasing speed. I am effectively arguing that all citizens consider speeding limits too low, thus speeding is preferred to not speeding. Thus, a driver will not reoffend if.

$$U(\text{Not reoffend}) > U(\text{Reofend})$$

$$\text{Formality}_i > \frac{F}{\text{Wealth}_i} + U(\bar{S} - S) \quad (1)$$

A very important element behind the model's intuition is the fact that fines in Bogotá (in Colombia, see footnote 6) are comparatively expensive. The fine is determined by the minimum monthly salary. For all drivers in my data a speeding fine is equivalent to half a monthly salary, or the equivalent of 80 working hours⁸. For

⁶ In the case of Colombia, for many years the only real cost was the inability to sell the car. Just in 2021 the legal right for the government to seize a driver's earning was granted. This has translated into a 20 percentage point increase in the numbers of tickets that are effectively paid in all relevant time frames.

⁷ The laws regulating traffic fines are national, and there is little local power to modify them.

⁸ I abstract here from early payment discounts. For all the infractions I am considering, a 50% discount is given if the ticket is pay before 12 days have passed from notification. The discount is 25% if is

comparison, in New York city a speeding fine ranges from 45 to 600 dollars⁹ (from 3 to 36 working hours). In Chicago the range is 35 to 100 dollars (from 2 to 6 hours)¹⁰. A higher price will have to be paid in the United Kingdom, where the fines go from £100 to £1,000 (or the equivalent of 8.7 to 87 work hours)¹¹. A city like Buenos Aires has a much closer price, with speeding fines ranging from \$217.800 to \$1.452.000 Argentinian pesos (from 0.73 to 4 times the monthly salary of 296.832 Argentinian pesos)¹². One could argue that there is a significant part of the population that cannot afford to pay such expensive fines and thus decides to remain in the informal sector; in practice expensive fines act as a barrier to formality¹³.

One could also think that the present value of legality is higher for wealthier drivers, so the effect of $Wealth_i$ is twofold: making the benefit higher while reducing the cost. This only complicates the model without changing the prediction. In any case, the model leads to a very straightforward hypothesis.

Hypothesis 1: The higher the $Wealth_i$ of a driver the higher its probability of not reoffending.

Note that inequality (1) without the term farthest to the right is equivalent to the decision to pay the fine. The decision whether to avoid a new fine is very similar to the decision of not having to pay a fine in the future. As my estimation wants to compare the random component that makes some drivers receive a fine and some not to receive a fine, I will not compare payment rates. But past payment rates could show if the logic of inequality (1) is supported by the data. As Figure 1 of the Appendix shows this is clearly the case; payment rates are linear on wealth, with the highest quintile having a payment rate that is close to 20 percentage points higher than the lower quintile. It is only indicative evidence since it could be the case that there is an omitted variable that determines both income and payment rates.

Law abiding behavior:

paid between 12 and 26%. Since this is common for all drivers, it does not change any of the model's predictions.

⁹ See Penalties in: <https://trafficsafety.ny.gov/penalties-speeding>. See: <https://dol.ny.gov/minimum-wage-0>

¹⁰ See: https://www.chicago.gov/city/en/depts/bacp/supp_info/minimumwageinformation.html
https://www.chicago.gov/city/en/depts/cdot/supp_info/children_s_safetyzoneprogramautomaticspeedenforcement/automated_speed_enforcementfrequentlyaskedquestions.html

¹¹ See: <https://www.evo.co.uk/advice/18773/uk-speeding-fines-2025-what-are-the-latest-penalties-for-drivers>

¹² See: <https://www.pagina12.com.ar/808914-aumentan-las-multas-de-transito-en-la-provincia-de-buenos-ai> <https://www.argentina.gob.ar/trabajo/consejodelsalario>

¹³ Not paying speeding fines does not allow for the vehicles to be immobilized. The most common reason for vehicles to be immobilized is not having the mandatory national insurance. Yet more than 60% of motorcycles on the road do not have the national insurance (See: <https://www.eltiempo.com/economia/sectores/el-62-5-de-las-motos-en-colombia-no-tiene-soat-estos-siniestros-generan-alto-costo-en-salud-3433613>). Informality on the road, on the country as a whole, is a real issue.

The second independent decision that a driver faces is whether to speed in the future on locations where there are no Cameras. I will call this decision change, since in those locations the chances of receiving a speeding ticket are essentially zero. Thus, the decision comes to only behavioral factors, which in my model has to do with the fairness of the specific fine received and the perception of the general actions of the state. Again, for simplicity I assume all citizens consider speeding limits too low. In this context fines act as a *wake up call* that drivers decide to follow depending on the perceived fairness of it.

Premise 3: After receiving a fine, the decision to speed on spots where Cameras are NOT located is the result of the comparison of the perception of the fairness of the previous fine with the perceived utility of speeding.

For simplicity I assume that drivers derive utility from feeling like good citizens and thus will adjust their behavior when the fines received are considered by them as fair. The fairness of the fine depends on two elements. The first is the relative price of the previous fine. The intuition is if citizens feel that the government is taking too much money out of them, they will feel the purpose of the fine is not correct behavior but increasing revenue. I will model this as a distance of the fine F with what each drivers would consider a fair fine F^* . For simplicity I will assume that F_i^* is an increasing function of $Wealth_i$. I will also assume that for all drivers $(F - F_i^*)^2 > 0$. The intuition behind is that no one considers a fine to be fair but the higher the income the fairer a fixed fine will be perceived¹⁴. The second element that defines fairness is the general perception of the state as an actor that is worried about safety on the road and not about increasing revenue. I will call this perception $Legitimacy_i$, I will assume that it is increasing in $Wealth_i$.

But I will assume that this relation has a peak, and that for a very high level of wealth there is low $Legitimacy_i$. With this non-linearity of fairness of the government action on income I am capturing the feeling that very rich people have that the State takes more from them than it should. Two elements help justify the intuition of the low $Legitimacy_i$ at the highest level of wealth. The first is a fact of the specific taxes on vehicles. It is a fact that proportionally more expensive vehicles pay higher taxes and higher premiums on the mandatory insurance. The second is shown in Figure 2 of the Appendix using survey data. While perceptions of corruption are decreasing on wealth, there is a jump on the very rich. This is true both for corruption of the mayor's office (for all the country) and for the police¹⁵ (for Bogotá).

In short, the comparison of utility of law-abiding behavior compared to the utility of speeding can be modeled as:

¹⁴ Once again, speeding fines are particularly rigid in this context. Not only is the amount to be paid always the same, it is the same for all levels of speeding.

¹⁵ Note that the national police give traffic tickets. In previous work (Ferro, 2024) I have shown that perceptions of corruption of the police make traffic ticket less deterrent.

$$U(\text{Change behavior}) > U(\text{Speeding})$$

$$\left(\frac{1}{(F - F_i^*)^2} \right) + \text{Legitimacy}_i > U(\bar{S} - S)$$

This also leads to a straightforward hypothesis.

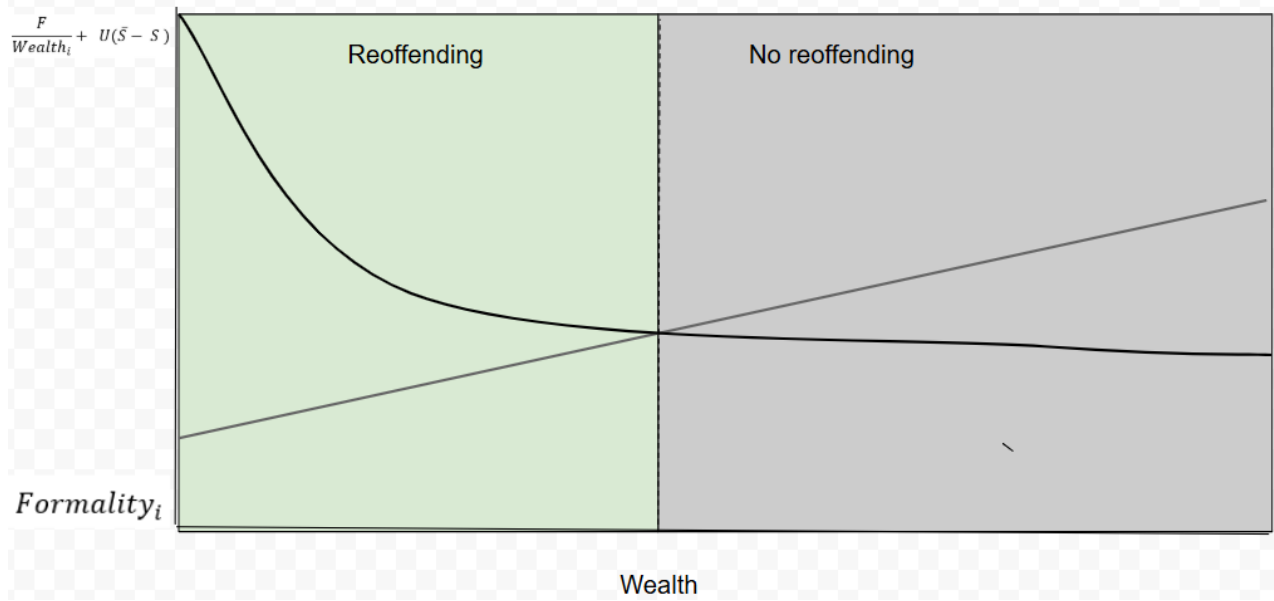
Hypothesis 2: In general, the higher the $Wealth_i$ of a driver the higher its probability of not speeding on spots where Cameras are NOT located. But for very low levels and very high levels of wealth, there might not be behavioral change.

The last sentence summarizes my intuition. For very low levels of $Wealth_i$, the optimal the fair amount of a fine F_i^* is very low, thus the first term of the summation, $\left(\frac{1}{(F - F_i^*)^2} \right)$, will be very low. If we add this to the hypothesis that this group has very low $Legitimacy_i$, since it is increasing in $Wealth_i$, the very poor might prefer to continue speeding after receiving a fine. The fine is so expensive for them that they won't even consider the wake up call.

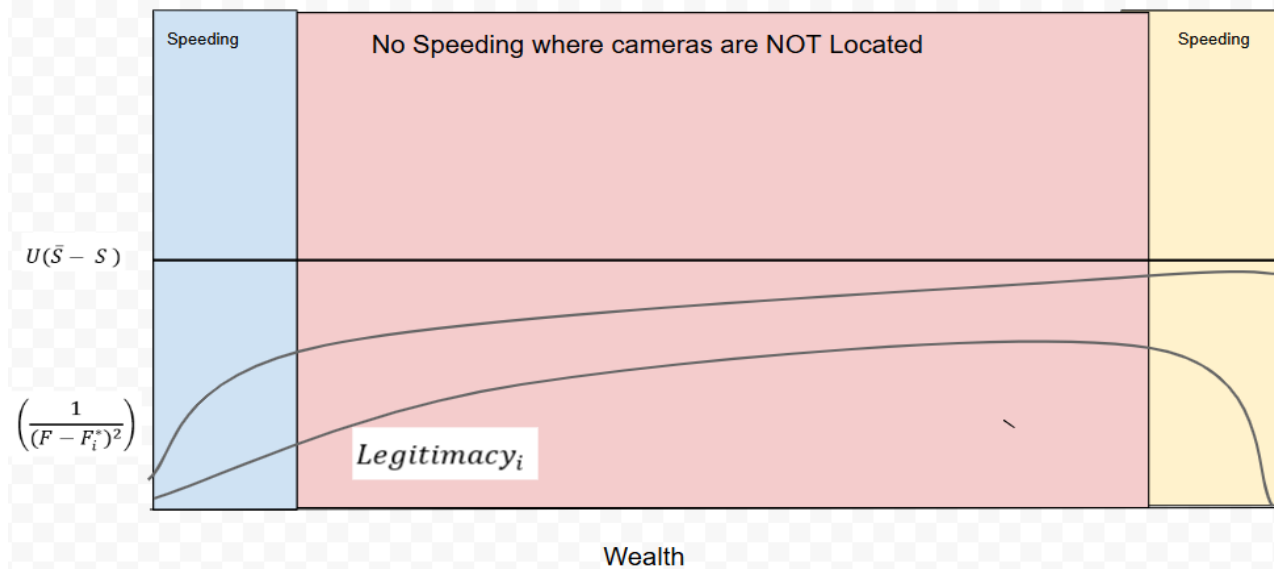
The intuition for the very high on $Wealth_i$ is similar. For them the first term is very the first term of the summation, $\left(\frac{1}{(F - F_i^*)^2} \right)$ is high by construction, since F_i^* , is the highest. But as I argued, their $Legitimacy_i$ is quite low. In short, this group perceives that even if the fine is little for them monetarily, it is just a manifestation of an illegitimate authority that already asks a lot from them. Thus, they might consider but not accept the wake up call.

Graph 1 summarizes both decisions and how they interact to create all four scenarios in terms of the effect of a ticket by the wealth of each driver.

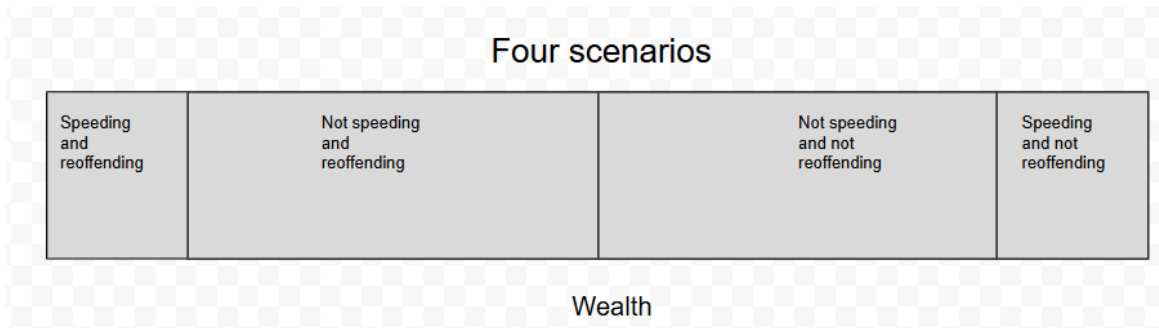
Reoffending



Behavioral change



Summary graph, with all four possible scenarios in terms of wealth:



The purpose of the model is to be flexible enough for all combinations of the two decisions. One should note, as I will explain later, that there is much more data for testing my Hypothesis 1 than my Hypothesis 2. I have observations for speed under the Cameras but I do not have speed measures outside of the Cameras. What I do have are accidents. I am effectively assuming that speeding outside of the location of the Cameras will result in accidents, and testing this relation too.

4. Estimation strategy.

My estimation derives from a fact that is unknown to drivers, that tickets are not given at the speed posted on the road. On the following I will call the legal limit posted on the road sign (50 km/h) as the posted speed limit, and the limit at which most tickets really given (55 km/h) as the real limit¹⁶.

This discontinuity allows for some exogeneous variation that can be exploited. In most of my applications I will estimate the following equation:

$$Result_i = f(z_i) + \tau D_i + \epsilon_i \quad (1)$$

Where $Result_i$ is one of my three behaviors of interest for vehicle i : a dummy that takes the value of 1 if the vehicle had at least one instance of excess speed¹⁷ in the six months after receiving a ticket and 0 otherwise, a dummy that takes the value of 1 if the vehicle had one accident in the six months after receiving a ticket and 0 otherwise, or the total number of instances of excess speed in the six months after receiving a ticket.

$f(z_i)$ is a polynomial of degree 2 (Gelman & Imbens, 2019) for the variable that decides treatment, the speed at which vehicle i is circulating under a Speed Camera. All vehicles in the sample exceed the posted speed limit, but only some exceed the real limit. This reduces selection and allows me to identify causality.

¹⁶ In the sample there is only one Camera that has 30 as the maximum speed. It only accounts for 4,460 observations (1.82%). Speeds are adjusted accordingly.

¹⁷ Since the real limit is unknown to drivers, a future excess speed is the one at above the posted limit; that is 50 km/h, not 55 km/h.

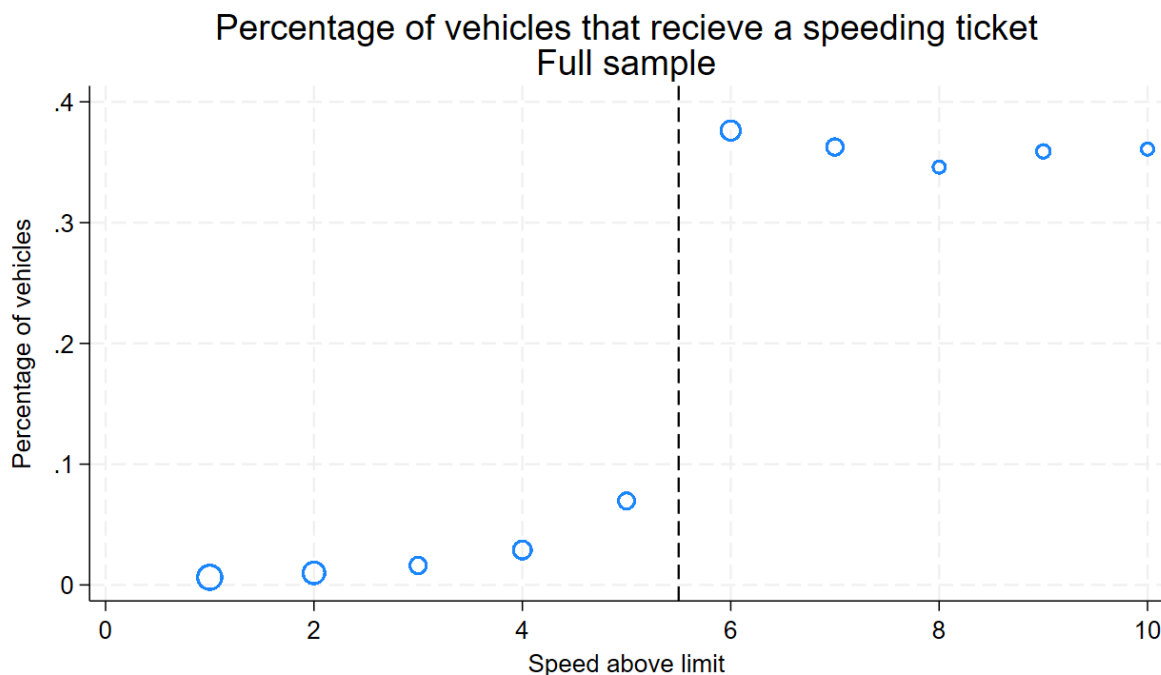
D_i is my treatment variable, that takes the value of 1 if the vehicle received a speeding ticket and 0 otherwise. The treatment is instrumented via the assignment, since this is a fuzzy design, and the discontinuity at the real speed limit is around an increase of 27 p.p. on the probability of receiving a ticket.

ϵ_i is an error term.

τ is my coefficient of interest. I will argue that it captures the causal effect of receiving a ticket in subsequent behavior. It is a local effect, since it only captures the effect of a ticket for vehicles around the cutoff, but those are the vast majority of observations in the sample¹⁸. For all estimations I use the `rdrobust` package in STATA with the option to ignore mass points, as the variable that assigns treatment is discrete. Three estimator, conventional, bias-corrected and robust will be shown.

4.1 Validation of the Fuzzy Regression Discontinuity Design: Existence of discontinuity:

Figure 1 shows that enforcement does work with the real limit, not the one posted on the street signs. It also shows that the discontinuity is not 0 to 1, so my strategy will be a fuzzy design. In short, even if the behavior of driving at 55 km/h is very similar to driving at 56 km/h, and both are above the legal limit, the latter has four times the probability of receiving a ticket than the former.



¹⁸ 80% of observations pass under Speed Cameras at speeds between 51 and 60 km/h.

Figure 1. Speed above limit is constructed as: (vehicle observed speed-speed limit).

It is important to show that this discontinuity also exists on the matched sample and that is similar for different values of vehicle prices. Figure 2 plots this discontinuity for all matched vehicles and for each quintile of vehicle price. In fact, the discontinuity is slightly bigger in this case across the board, but particularly for higher quintiles.

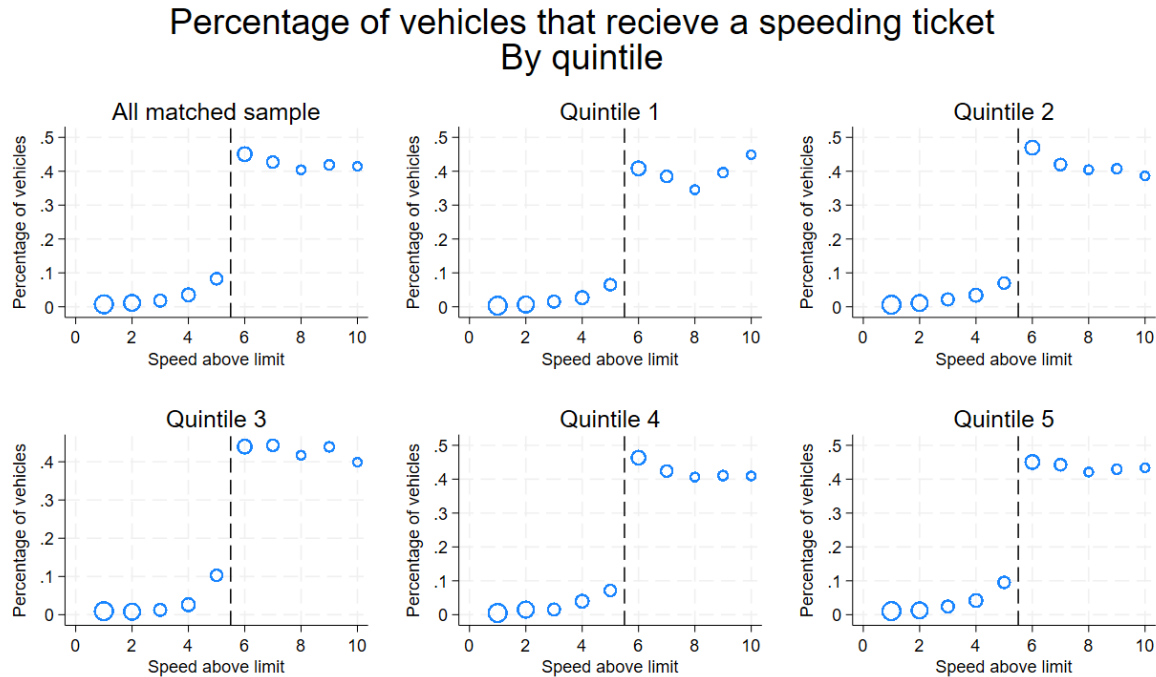


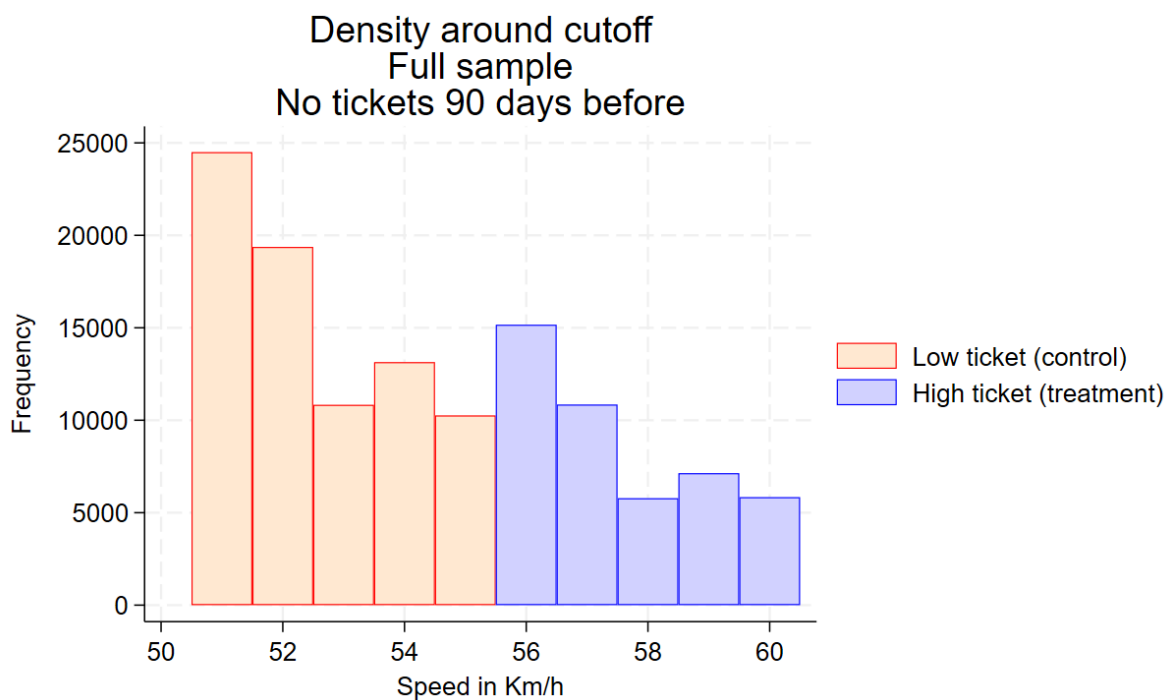
Figure 2. Speed above limit is constructed as: (vehicle observed speed-speed limit).

A very relevant question for local public policy is whether this deterrent effect of tickets varies by vehicle type. This is caused by motorcycles being a growing percentage of road injuries and fatalities both in Bogotá and in all Colombian cities, as in similar middle income countries. Graphs 3 and 4 of the Appendix show this for the full and matched sample. It can be seen that the discontinuity is smaller for motorcycles, and that this is considerably smaller in the case of the matched sample. This is less a threat for the identification strategy and more of a fact to have in mind when interpreting results.

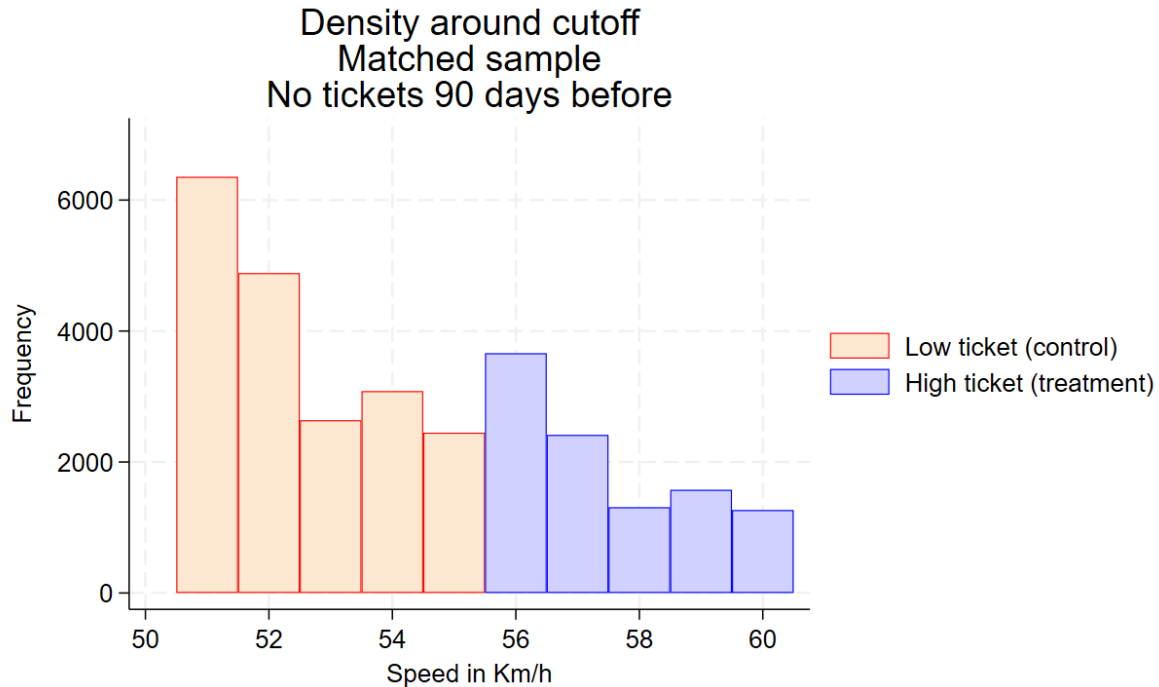
4.2 No manipulation.

Now that I have shown that there is a discontinuity at cutoff, I need to show that there is no manipulation. In short, drivers are not aware of this fact. Since the estimation wants to separate the effect of a possible ticket from any other change, I will only use the vehicles that have not received a ticket in the 90 days before passing under a Camera exceeding the legal speed limit. Frequency is shown in Graph 5. In the Appendix the same distribution is shown for all vehicles, independent of receiving or

not receiving a ticket in the prior 90 days. It seems that drivers are not selecting themselves into not receiving a ticket.

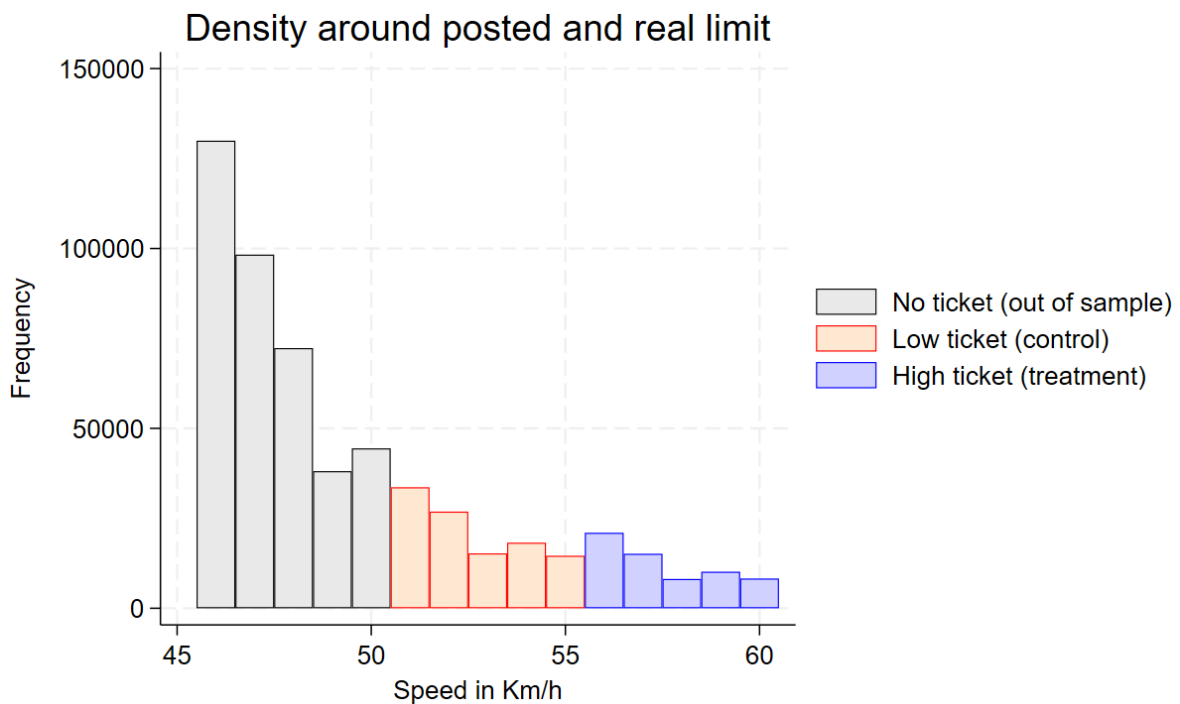


Graph 3. Frequency by condition. Full and sample.



Graph 4. Frequency by condition. Matched sample.

Another way to show that drivers are not selecting themselves into not receiving a ticket is to see what happens at the posted limit. One can see that there is bunching to the left of the posted limit, but to the right of the real limit.



Graph 5. Discontinuity around posted and real limit.

4.3 Smoothness on observables.

4.3.1 Full sample:

Another element I need to evaluate the validity of the fuzzy regression discontinuity design is to measure the smoothness of observable variables of the vehicles involved. I will do this first on the full sample, and then on the matched sample. For all estimations I use a polynomial of degree 2, and a bandwidth of 4.5 below and 10.5 above the cutoff (vehicles driving between 51 and 65 Km/h) with a triangular kernel. This leaves me with 136533 effective observations (76528 below and 60005 above the cutoff), of vehicles that only exceed the speed limit once during the sample period and did not receive a speeding ticket on the 90 days before doing so.

The first variable I explore is the hour of the day at which the ticket is received, a possible indications that drivers are different in the treatment and control groups. The same selection can be analyzed with day of the week as a continuous variable or a dummy if the ticket was given during the weekend. Since in Colombia, plates for motorcycles have a different configuration than those of other vehicles¹⁹, I can observe the type of vehicle involved. I use a dummy that takes the value of 1 if the vehicle involved is a motorcycle. As Table 6 shows, the only difference that one can observe is by vehicle type, and it is still small.

Smoothnes of observables Full sample

	(1) Hour	(2) Day	(3) Weekend	(4) Motorcycle
Conventional	-0.0401 (-0.19)	0.101 (0.79)	-0.00233 (-0.08)	-0.0660** (-2.27)
Bias-corrected	0.211 (0.98)	0.0923 (0.72)	-0.00149 (-0.05)	-0.0737** (-2.53)
Robust	0.211 (0.65)	0.0923 (0.48)	-0.00149 (-0.03)	-0.0737* (-1.68)
N	136533	136533	136533	136533
Mean	11.80	4.156	0.350	0.341

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

¹⁹ Motorcycle plates end in a letter, whether all other vehicle's plates end in a number.

Table 6. Smoothness on observables. Full sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise.

4.3.2 Matched sample:

In this case the total number of observations is 62288 (36383 below and 25905 above the cutoff. Just like with the full sample, most of the variables are smooth at the cutoff. It is also the case that there are fewer motorcycles are the cutoff, but one should note the big difference in percentages between the two samples. Whereas in the Full Sample 34% of the vehicles are motorcycles, this percentage is only 7.5% in the matched sample. This reinforces my argument about the Matched sample not being a representative part of the Full Sample.

Note that there is no significant difference in terms of price.

Smoothnes of observables Matched sample

	(1) Hour	(2) Day	(3) Weekend	(4) Motorcycle	(5) Price
Conventional	-0.317 (-1.22)	-0.00624 (-0.04)	-0.0147 (-0.40)	-0.0172 (-0.87)	-0.831 (-0.20)
Bias-corrected	-0.311 (-1.20)	0.113 (0.70)	0.000722 (0.02)	-0.0468** (-2.37)	-4.353 (-1.07)
Robust	-0.311 (-0.79)	0.113 (0.46)	0.000722 (0.01)	-0.0468 (-1.58)	-4.353 (-0.71)
N	62288	62288	62288	62288	62288
Mean	11.94	4.312	0.399	0.0755	56.38

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 7. Smoothness on observables. Full sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise.

5. General results.

In this section I show all the results. In section 5.1 I presents all results on reoffending – speeding. In Section 5.2 I presents results on Behavioral change – accidents. Section 5.3 is dedicated to the relative price analysis.

5.1 Reoffending – Speeding:

5.1.1. Full sample:

Table 8 presents the results for equation (1) with different dependent variables related to speeding for a period of six month after a vehicle is observed speeding under a Camera. Speeding relates to the fact that a vehicle was observed at a speed higher than 50 km/h at least once. Serious speeding (or Speeding>10) refers to the fact that a vehicle was observed at a speed higher than 60 km/h at least once.

This are the most important results of the paper. The results point at ticket having a causal effect of reducing all types of speeding behavior. This reduction is significant in magnitude. It is 64% for ever having a speeding infraction to 94% reduction for having a serious one.

	(1) Speeding	(2) TimesSpeeding	(3) Speeding>10	(4) TimesSpeeding>10
Conventional	-0.114*** (-4.47)	-0.178*** (-3.21)	-0.0331** (-2.34)	-0.0410* (-1.88)
Bias-corrected	-0.145*** (-5.68)	-0.238*** (-4.29)	-0.0538*** (-3.80)	-0.0664*** (-3.05)
Robust	-0.145*** (-3.77)	-0.238*** (-2.85)	-0.0538** (-2.53)	-0.0664** (-2.05)
N	136533	136533	136533	136533
Mean	0.225	0.371	0.0567	0.0743

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 8. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

5.1.2 Full sample, by vehicle type:

A central question for policy makers is the effectiveness of speeding tickets and other measures for speed control in the face of the growing number of motorcycles in middle income countries. For simplicity I only make a dummy that says if a vehicle is a motorcycle (=1) or not (=0), and I will call not-motorcycles as cars. As it is shown in the Appendix, even if the discontinuity is much smaller for motorcycles, it is still present for both values of my dummy variable.

As Table 9 and 10 show, the general effect found on speeding is really only captured by cars. None of the coefficients for motorcycles are significant.

Vehicle type:

Cars

	(1) Speeding	(2) TimesSpeeding	(3) Speeding>10	(4) TimesSpeeding>10
Conventional	-0.152*** (-4.64)	-0.229*** (-3.13)	-0.0359** (-2.04)	-0.0464* (-1.70)
Bias-corrected	-0.207*** (-6.31)	-0.313*** (-4.28)	-0.0653*** (-3.71)	-0.0786*** (-2.88)
Robust	-0.207*** (-4.19)	-0.313*** (-2.85)	-0.0653** (-2.48)	-0.0786* (-1.94)
N	89037	89037	89037	89037
Mean	0.249	0.419	0.0581	0.0759

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 9. Reoffending. Full sample: Cars. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

Motorcycles

	(1) Speeding	(2) TimesSpeed~g	(3) Speeding>10	(4) TimesSpee~10
Conventional	-0.0532 (-1.34)	-0.104 (-1.30)	-0.0265 (-1.11)	-0.0299 (-0.83)
Bias-corre~d	-0.0411 (-1.03)	-0.115 (-1.44)	-0.0324 (-1.36)	-0.0449 (-1.24)
Robust	-0.0411 (-0.68)	-0.115 (-0.95)	-0.0324 (-0.90)	-0.0449 (-0.83)
N	47496	47496	47496	47496
Mean	0.170	0.266	0.0526	0.0688

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 10. Reoffending. Full sample: Motorcycles. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had a at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had a at least one serious speeding.

5.1.3. Matched sample:

I can also do this analysis in the matched sample. As Table 11 shows the results are essentially the same, a general effect of reduction in all types of speeding. Note also that the result is slightly less robust than in the general sample.

Matched sample:

	(1) Speeding	(2) TimesSpeed~g	(3) Speeding>10	(4) TimesSpee~10
Conventional	-0.103*** (-3.15)	-0.145** (-2.18)	-0.0153 (-0.88)	-0.0161 (-0.67)
Bias-corre~d	-0.125*** (-3.82)	-0.168** (-2.53)	-0.0461*** (-2.67)	-0.0531** (-2.20)
Robust	-0.125** (-2.53)	-0.168* (-1.69)	-0.0461* (-1.78)	-0.0531 (-1.48)
N	62288	62288	62288	62288
Mean	0.252	0.395	0.0566	0.0687

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 11. Reoffending. Matched sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

5.1.4 Matched sample, by vehicle type:

Tables 12 and 13 should be read with caution, since most of the matched sample is comprised of cars. Having said that, the tendency is exactly the same; cars explain most of the results on speeding.

Cars

	(1) Speeding	(2) TimesSpeed~g	(3) Speeding>10	(4) TimesSpee~10
Conventional	-0.107*** (-3.20)	-0.143** (-2.11)	-0.0146 (-0.84)	-0.00952 (-0.40)
Bias-corre~d	-0.133*** (-3.99)	-0.171** (-2.53)	-0.0472*** (-2.71)	-0.0466* (-1.94)
Robust	-0.133*** (-2.65)	-0.171* (-1.68)	-0.0472* (-1.81)	-0.0466 (-1.30)
N	57564	57564	57564	57564
Mean	0.267	0.419	0.0577	0.0697

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 12. Reoffending. Full sample: Motorcycles. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

Motorcycles

	(1) Speeding	(2) TimesSpeed~g	(3) Speeding>10	(4) TimesSpee~10
Conventional	-0.0534 (-0.36)	-0.211 (-0.66)	-0.0222 (-0.24)	-0.127 (-0.82)
Bias-corre~d	-0.0300 (-0.20)	-0.215 (-0.68)	-0.0153 (-0.16)	-0.161 (-1.05)
Robust	-0.0300 (-0.13)	-0.215 (-0.44)	-0.0153 (-0.11)	-0.161 (-0.70)
N	4724	4724	4724	4724
Mean	0.169	0.274	0.0559	0.0749

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 13. Reoffending. Full sample: Motorcycles. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

5.2 Behavioral change – accidents:

5.2.1 Full sample:

Table 14 presents the results of equation (1) when the dependent variable measures accidents. Neither a dummy for having an accident (column 1), nor the continuous variable for total of accidents (column 2), nor the dummy for accident with at least one fatality (column 3) are significant. Nor is it useful to separate the sample into cars and motorcycles.

	(1) Anyaccident	(2) TotalAccid~t	(3) Deadlyacci~t	(4) Cars	(5) Motorcycles
Conventional	-0.00464 (-1.01)	-0.00469 (-1.01)	-0.000644 (-0.78)	-0.00710 (-1.25)	0.000196 (0.02)
Bias-corrected	-0.00558 (-1.21)	-0.00520 (-1.12)	-0.00118 (-1.43)	-0.00589 (-1.03)	-0.00504 (-0.64)
Robust	-0.00558 (-0.81)	-0.00520 (-0.75)	-0.00118 (-0.99)	-0.00589 (-0.69)	-0.00504 (-0.43)
N	136533	136533	136533	89037	47496
Mean	0.00544	0.00551	0.000193	0.00538	0.00555

t statistics in parentheses

* p<0.05, ** p<0.01, *** p<0.001

Table 14. Behavioral change. Full sample. Column 1 (Anyaccident) is a dummy that takes the value of 1 if the vehicle had at least one accident in the following months, and 0 otherwise. Column 2 (TotalAccidents) is a continuous variable; the total of times a vehicle had an accident. Column 3 (DeadlyAccident) is a dummy that takes the value of 1 if the vehicle had at least one accident with a fatality in the following months, and 0 otherwise. Column 4 (Cars) is the same as Column 1 restricting the sample to cars. Column 5 (Motorcycles) does the same for motorcycles.

A possible objection to these results could be the fact that the estimation is underpowered. I should not that the same number of observations produced a very credible result for speeding. I should also point out that there are 724 accidents in my estimation sample. These are around 6% of the total 11922 total accidents that occurred in the period of study. One could extend the sample some additional months to obtain a bigger number of accidents. When widening the time period, even if the total of accidents in the sample increases to 892 (one additional month) or 928 (two months), the result (not reported) does not change.

5.2.1 Matched sample:

The result holds if one looks at the Matched sample. As can be seen in Table 15, results are essentially the same; although point estimates are negative, they are far from significant.

	(1) Anyaccident	(2) TotalAccident	(3) Deadlyaccident
Conventional	-0.00830 (-1.35)	-0.00837 (-1.36)	-0.000231 (-0.22)
Bias-corrected	-0.00196 (-0.32)	-0.00175 (-0.28)	-0.000481 (-0.46)
Robust	-0.00196 (-0.21)	-0.00175 (-0.19)	-0.000481 (-0.31)
N	62288	62288	62288
Mean	0.00630	0.00638	0.000178

t statistics in parentheses

* p<0.10, * p<0.05, * p<0.01

Table 15. Behavioral change. Matched sample. Column 1 (Anyaccident) is a dummy that takes the value of 1 if the vehicle had at least one accident in the following months, and 0 otherwise. Column 2 (TotalAccidents) is a continuous variable, the total of times a vehicle had an accident. Column 3 (DeadlyAccident) is a dummy that takes the value of 1 if the vehicle had at least one accident with a fatality in the following months, and 0 otherwise.

5.2.2. Results by relative price.

As I have shown before, the results are only significant for dependent variables related to reoffending on speeding behaviors. Thus, I will only show the heterogeneous effect for those. Table 16 reproduces equation (1) with the dependent variable of a dummy of speeding (column 1 of table 8). Table 17 does the same for the dummy of serious speeding (column 3 of table 8).

Results are similar, but stronger for speeding. The effect is concentrated on the higher quintiles of the distribution.

Matched Sample Quintiles

	(1) Quintile1	(2) Quintile2	(3) Quintile3	(4) Quintile4	(5) Quintile5
Conventional	-0.0799 (-0.86)	-0.102 (-1.51)	-0.105 (-1.29)	-0.119* (-1.96)	-0.0889 (-1.32)
Bias-corrected	-0.0676 (-0.73)	-0.0538 (-0.80)	-0.151* (-1.85)	-0.161*** (-2.65)	-0.141** (-2.09)
Robust	-0.0676 (-0.48)	-0.0538 (-0.53)	-0.151 (-1.22)	-0.161* (-1.75)	-0.141 (-1.36)
N	7078	11355	11993	15750	16112
Mean	0.168	0.241	0.250	0.269	0.282

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 16. Reoffending by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise.

Matched Sample Quintiles

	(1) Quintile1	(2) Quintile2	(3) Quintile3	(4) Quintile4	(5) Quintile5
Conventional	0.0185 (0.38)	-0.0418 (-1.18)	0.0268 (0.63)	-0.0152 (-0.48)	-0.0324 (-0.86)
Bias-corrected	0.0289 (0.59)	-0.0639* (-1.81)	0.00780 (0.19)	-0.0446 (-1.42)	-0.0942** (-2.51)
Robust	0.0289 (0.39)	-0.0639 (-1.23)	0.00780 (0.12)	-0.0446 (-0.95)	-0.0942 (-1.64)
N	7078	11355	11993	15750	16112
Mean	0.0424	0.0508	0.0535	0.0554	0.0705

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 17. Reoffending by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise.

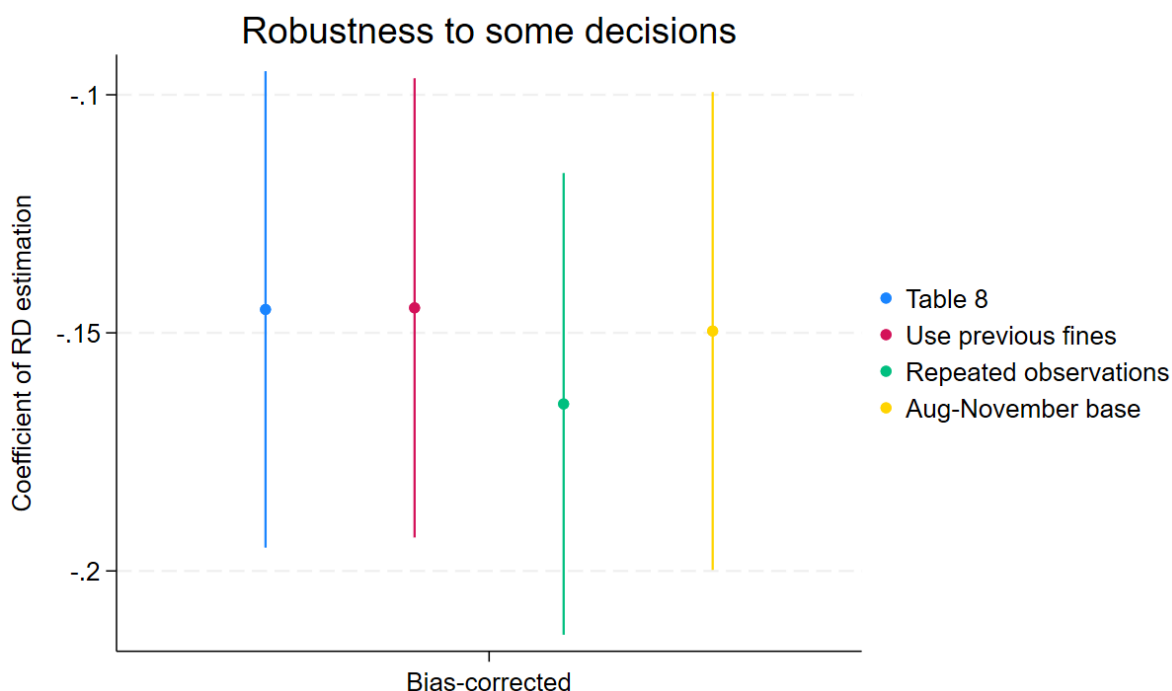
6. Robustness checks.

In this section I introduce some further evidence that my results are credible. First, I present alternative estimations to show consistency. Second, I use an alternative method of estimating my results: an IV-judge type of estimation using data on the police agent in charge of verifying each speeding ticket.

6.1 Robustness to some decisions:

Graph 18 shows a comparison between the coefficient found on column 1 of Table 8 and the coefficient found with different decisions. In particular, whether to include vehicles that had a previous fine, to use vehicles that pass more than once under a Camera and to have a bigger time for selecting my sample.

The coefficient is very stable across all types of estimations, and bigger for the case in which vehicles that passed more than once are included. This is discussed in bigger detail in section 2 of the Appendix.

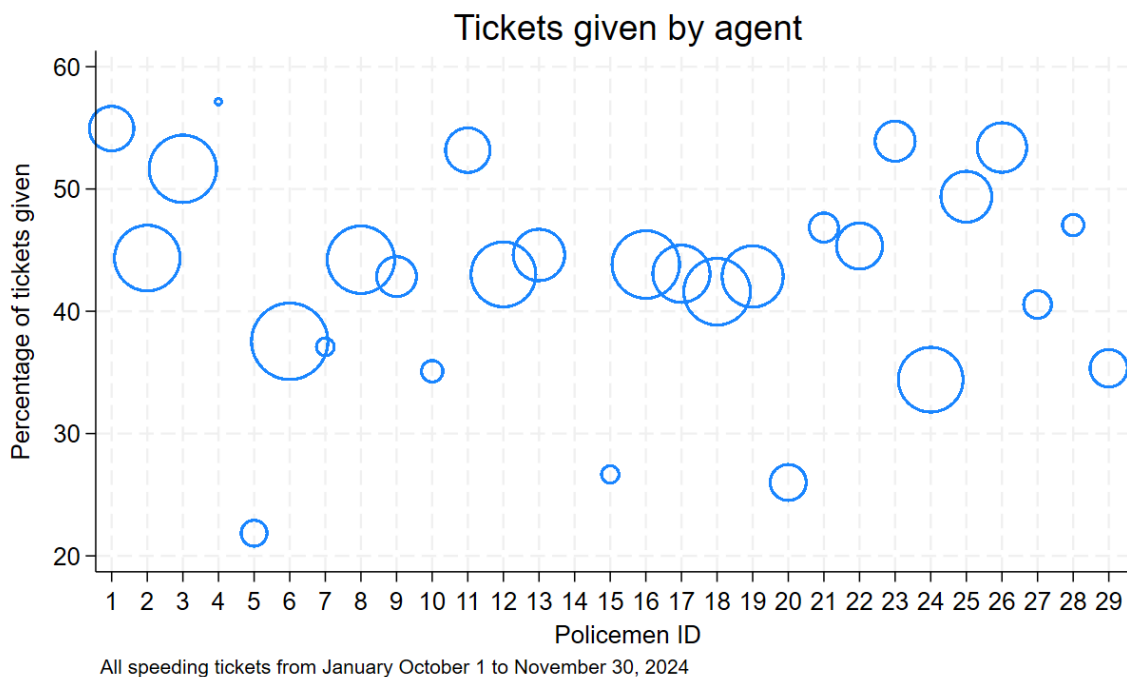


Graph 18. “Use previous fines” uses the same estimation, this time including vehicles that have had a ticket in the previous six months. “Repeated observations” shows the coefficient using Vehicles that passed more than once, as it is further discussed in section 2 of the Appendix. “Aug-November” base makes the database bigger, by including al vehicles from August 16 to November 15, 2024.

6.2 An IV-judge using police officer manual validation:

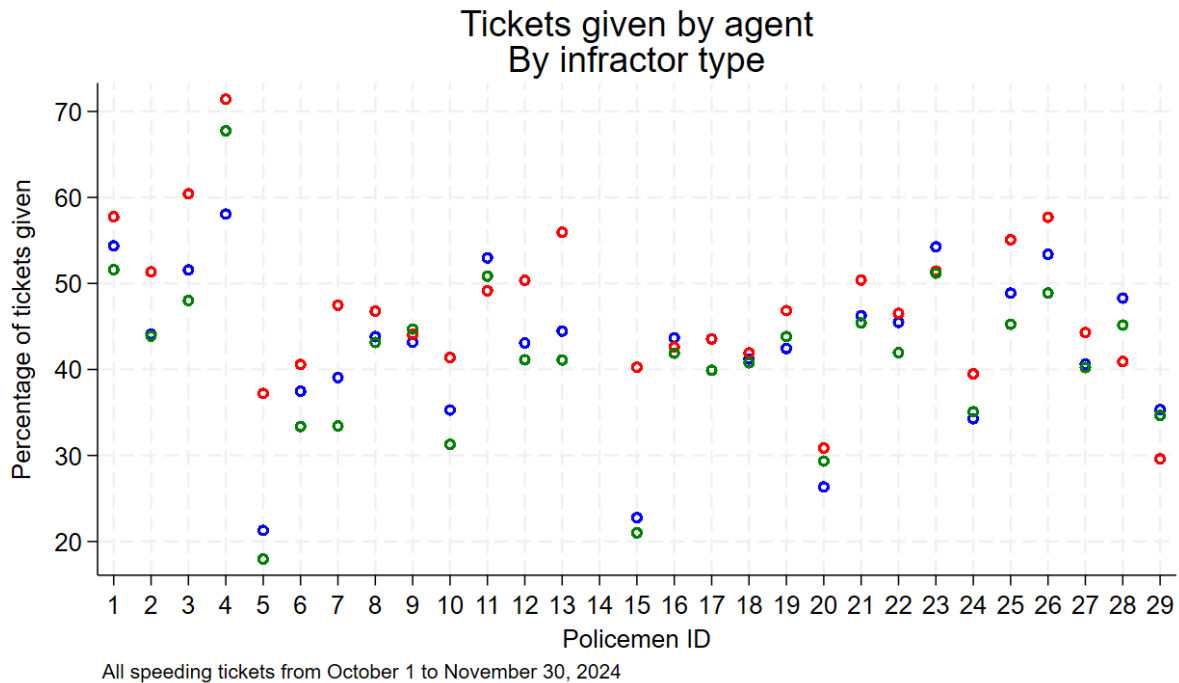
According to Colombian law all speeding tickets must be subject to human verification of their validity. This has to do mainly with the fact that the technology for taking pictures of vehicles while speeding is not perfect, and some pictures are blurry. This task is done by trained police officers that must evaluate whether a specific picture should transform into a ticket. The pictures assigned to each policemen are random, and they have no way to refuse their verification.

I have data for the identification of all pictures of vehicles that were circulating above 56 km/h during the months of October and November. I also can see which agent was in charge of reviewing each picture. As Graph 19 shows, there is considerable heterogeneity in the percentage of pictures that each agent transforms into a ticket.



Graph 19. Policeman ID is a number to identify each police officer. The size of the circle is proportional to the number of pictures reviewed by each agent.

Graph 19 indicates that there might be some randomness associated to the Policemen in charge. For this to transform into a credible instrumental variable one would want to verify that these different rates are not related to any observable characteristic of the ticket, or the vehicle involved. Graph 20 shows that the strictness is stable for all Policemen across at least two meaningful variables, whether a vehicle is a motorcycle and those vehicles that drive faster than 60 km/h.



Graph 19. Policeman ID is a number to identify each police officer. Blue dots correspond to all pictures, red dots to pictures of motorcycles and green to people under serious speeding (40% of this subsample).

Both previous graph give evidence to use the random assignment of the Policemen in charge of reviewing each picture as an instrument of the ticket. I will then capture the part of receiving a ticket that is due to the randomness of the assigned policemen. I also must emphasize that this is a different sample, consisting mostly of drivers that decide to exceed speeding limits seriously. I should also note that in this case I am not finding a local effect, but the general effect on those that decide to exceed the limit by a significant amount of kilometers.

Table 21 shows the result of an estimation with the same dependent variables as Table 8. As one can clearly see, the significance and sign of all the estimation is similar, but magnitudes are considerably smaller. While the effect of a speeding ticket was a 64% reduction in the change of reoffending, in this case the effect is only 25%. This surely is caused by these drivers deciding to disobey speeding limits by a bigger amount than the other sample.

	(1) Speeding	(2) TimesSpeed~g	(3) Speeding>10	(4) TimesSpee~10
comparendo	-0.0485*** (-2.85)	-0.105*** (-3.17)	-0.0194** (-2.17)	-0.0246* (-1.94)
N	132668	132668	132668	132668
Mean	0.196	0.310	0.0470	0.0604

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 21. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding. This table is comparable to Table 8.

The results on Table 21 are in line with the main finding of this paper. One could also use this sample to verify the second finding: effects being explained only by cars, not motorcycles. Table 22 presents the same estimation. Note that in this case the effect is more precisely estimated for cars than motorcycles, but there is some for the latter. Once again, this does not cast doubt on previous findings but shows that serious speeders are more similar between them, independent of the type of vehicle they drive.

Cars

	(1) Speeding	(2) TimesSpeed~g	(3) Speeding>10	(4) TimesSpee~10
comparendo	-0.0487** (-2.48)	-0.117*** (-3.05)	-0.0159 (-1.60)	-0.0236* (-1.71)
N	94536	94536	94536	94536
Mean	0.209	0.332	0.0451	0.0572

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Motorcycles

	(1) Speeding	(2) TimesSpeed~g	(3) Speeding>10	(4) TimesSpee~10
comparendo	-0.0453* (-1.69)	-0.0434 (-0.88)	-0.0273* (-1.72)	-0.0280 (-1.23)
N	38132	38132	38132	38132
Mean	0.163	0.252	0.0519	0.0691

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 21. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding. This table is comparable to Table 9 and 10.

One could use this estimation strategy to verify the other two findings. As Table 22 shows there is also a null effect on all dependent variables related to the chance of being involved in an accident. In this case the coefficients are positive, not like the previous ones negative but not significant.

	(1) Anyaccident	(2) TotalAccid~t	(3) Deadlyacci~t	(4) Cars	(5) Motorcycles
comparendo	0.000223 (0.09)	0.000417 (0.17)	0.000590 (1.24)	0.000807 (0.31)	-0.00439 (-0.92)
N	132668	132668	132668	94536	38132
Mean	0.00377	0.00380	0.000145	0.00362	0.00419

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 22. Behavioral change. Full sample. Column 1 (Anyaccident) is a dummy that takes the value of 1 if the vehicle had at least one accident in the following months, and 0 otherwise. Column 2 (TotalAccidents) is a continuous variable; the total of times a vehicle had an accident. Column 3 (DeadlyAccident) is a dummy that takes the value of 1 if the vehicle had at least one accident with a fatality in the following months, and 0 otherwise. Column 4 (Cars) is the same as Column 1 restricting the sample to cars. Column 5 (Motorcycles) does the same for motorcycles. This table is comparable to Table 14.

The final finding one could verify has to do with the heterogeneous effect by vehicle price. As Table 23 shows the effect of a ticket on speeding behavior is in this case is also concentrated on the higher quintiles, even if unlike in the other

estimations the effect is also present at quintile 2, although in a smaller magnitude and significance levels.

	(1) MatchedSample	(2) Quintile1	(3) Quintile2	(4) Quintile3	(5) Quintile4	(6) Quintile5
comparing	-0.0828*** (-3.45)	-0.0936 (-1.26)	0.102* (1.93)	-0.0962** (-2.16)	-0.136*** (-2.88)	-0.129*** (-2.77)
N	48230	3562	8772	10223	12150	13523
Mean	0.234	0.150	0.210	0.227	0.246	0.266

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

In summary, although the sample for the previous sections is different from this one, nor are the estimation strategies capturing the same causal effect, one can safely say that these results give more credibility to the causal effects found. All evidence points in the same direction.

7. Discussion and possible policy recommendations.

This paper can be summarized in one sentence: speeding tickets causally reduce speeding but seem not to have an effect in reducing accidents. If one wants to be more specific, one could add that the reduction of speeding is concentrated both in lesser speeding infractions and in automobiles. This main finding is true both for the Full sample and for the Matched sample of vehicles that are registered in Bogotá.

I would argue that the main interpretation of this finding is that the presumed causal relation between speed and accidents is less important than assumed. If speeding tickets have a significant and negative effect on speeding behavior but they do not have any significant effect on accidents, then speed is not a significant cause for accidents. Other interventions are needed if a city like Bogotá (and other in middle income countries) want to face the public health challenge of motorcycles.

The fact that the desired effects of the policy are concentrated on vehicles, and absent in motorcycles has important policy implications. This might mean that speeding tickets are not the right tool for the changing face of the public health challenge of road fatalities that some cities are facing.

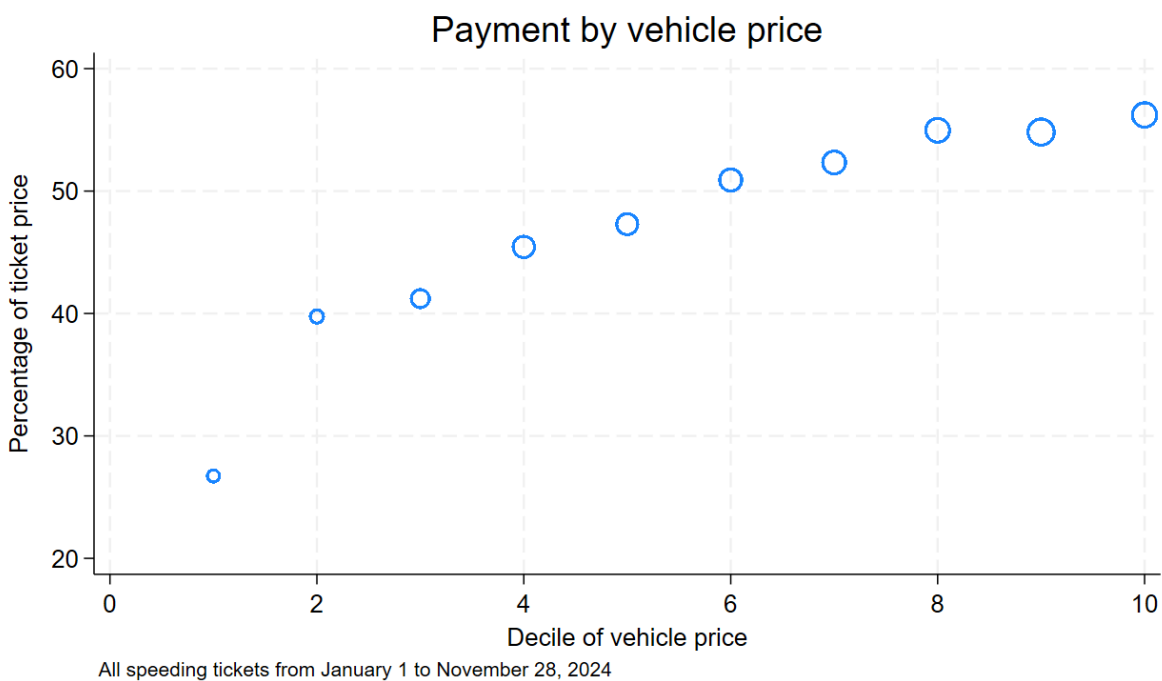
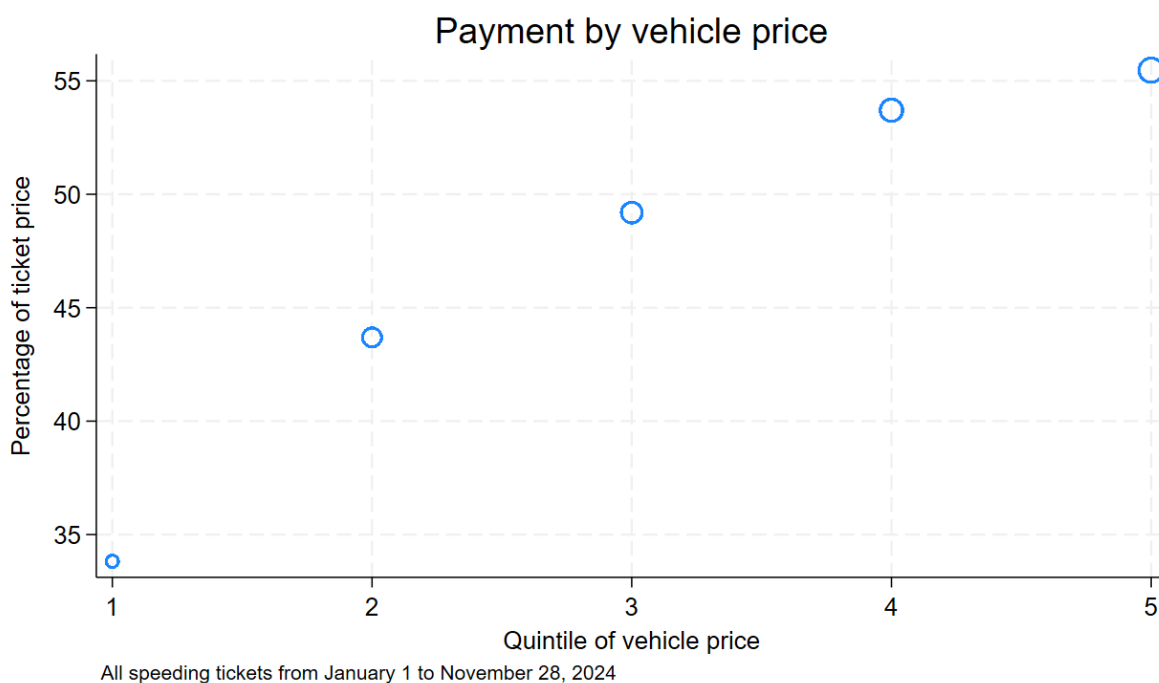
The last finding has to do with the effects on speeding being concentrated on the higher quintiles of the distribution. Coupled with the existing evidence that higher quintiles have higher payment rates, this might suggest that speeding tickets might have a better effect if they were adjusted, to a lower value, for the less wealthy drivers. Tickets might be too expensive for the majority of drivers, and according to my behavioral model can become a barrier to formality for the less wealthy.

8. References.

- Bertoli, P., & Grembi, V. (2021). The political cycle of road traffic accidents. *Journal of Health Economics*, 76. <https://doi.org/10.1016/j.jhealeco.2021.102435>
- Bing, L., Goldstein, R., Ho, H., Pager, D., & Western, B. (2024). *The long-term impact of debt relief for indigent defendants in a misdemeanor court*. <https://doi.org/10.1073/pnas>
- De Angelo, G., & Hansen, B. (2014). Life and Death in the Fast Lane: Police Enforcement and Traffic Fatalities. *American Economic Journal: Economic Policy*, 6(2), 231–257.
- Dušek, L., & Traxler, C. (2022). Learning from Law Enforcement. *Journal of the European Economic Association*, 20(2), 739–777. <https://doi.org/10.1093/jeea/jvab037>
- Finlay, K., Gross, M., Lieberman, C., Luh, E., & Mueller-Smith, M. (2024). The Impact of Criminal Financial Sanctions: A Multistate Analysis of Survey and Administrative Data. *American Economic Review: Insights*, 6(4), 490–508. <https://doi.org/10.1257/aeri.20230413>
- Fry, J. M., & Farrell, L. (2023). Road accidents: unexpected costs of stock market movements. *Oxford Economic Papers*, 75(1), 233–255. <https://doi.org/10.1093/oep/gpac016>
- Gelman, A., & Imbens, G. (2019). Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs. *Journal of Business & Economic Statistics*, 37(3).
- Giles, T. (2023). *The Government Revenue, Recidivism, and Financial Health Effects of Criminal Fines and Fees*. <https://ssrn.com/abstract=4568724> or <http://dx.doi.org/10.2139/ssrn.4568724>
- Giulietti, C., Tonin, M., & Vlassopoulos, M. (2020). When the market drives you crazy: Stock market returns and fatal car accidents. *Journal of Health Economics*, 70. <https://doi.org/10.1016/j.jhealeco.2019.102245>
- Goncalves, F., & Mello, S. (2024). *Police Discretion and Public Safety*. <https://www.fmgoncalves.com/s/draft-241112.pdf>
- Hansen, B. (2015). Punishment and deterrence: Evidence from drunk driving. In *American Economic Review* (Vol. 105, Issue 4, pp. 1581–1617). American Economic Association. <https://doi.org/10.1257/aer.20130189>
- Kaila, M. (2024). *How Do People React to Income-Based Fines? Evidence from Speeding Tickets Discontinuities*. https://marttikaila.com/assets/docs/jmp_kaila.pdf
- Lu, F., Zhang, J., & Perloff, J. M. (2016). General and specific information in deterring traffic violations: Evidence from a randomized experiment. *Journal of Economic Behavior and Organization*, 123, 97–107. <https://doi.org/10.1016/j.jebo.2015.12.009>

- Luca, D. L. (2015). Do Traffic Tickets Reduce Motor Vehicle Accidents? Evidence from a Natural Experiment. *Journal of Policy Analysis and Management*, 34(1), 85–106.
<https://doi.org/10.1002/pam.21798>
- Makowsky, M. D., & Stratmann, T. (2011). More Tickets, Fewer Accidents: How Cash-Strapped Towns Make for Safer Roads. In *Journal of Law & Economics* (Vol. 54, Issue 4).
<https://heinonline.org/HOL/License>
- Norris, S., & Rose, E. K. (2024). Laffer’s day in court: The revenue effects of criminal justice fees and fines. *Journal of Public Economics*, 240, 105249.
<https://doi.org/10.1016/j.jpubeco.2024.105249>
- Philippe, A. (2024). Learning by Offending: How Do Criminals Learn about Criminal Law? *American Economic Journal: Economic Policy*, 16(3), 27–60.
<https://doi.org/10.1257/pol.20210378>
- Traxler, C., & Dušek, L. (2023). Fines, nonpayment, and revenues: evidence from speeding tickets. *The Journal of Law, Economics, and Organization*.
<https://doi.org/10.1093/jleo/ewad025>
- Zhang, J., Perloff, J. M., & Lu, F. (2020). Informing and inquiring: Experimental evidence on reducing traffic violations. *Journal of Development Economics*, 147.
<https://doi.org/10.1016/j.jdeveco.2020.102534>

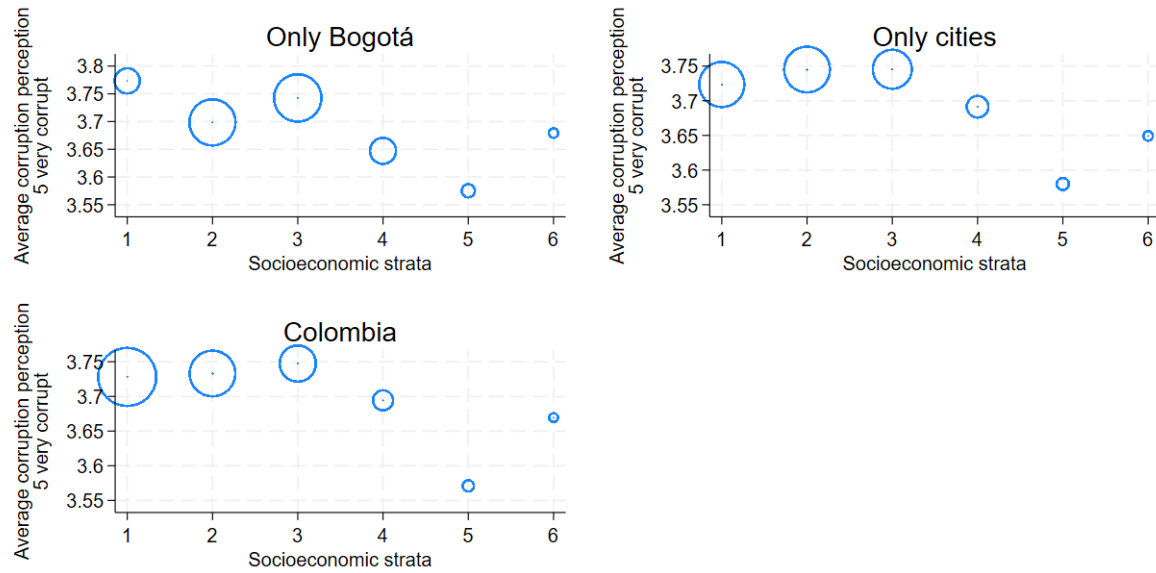
Appendix:



repetidos	Freq.	Percent	Cum.
1	158,703	87.44	87.44
2	17,731	9.77	97.21
3	3,460	1.91	99.12
4	934	0.51	99.64
5	371	0.20	99.84
6	142	0.08	99.92
7	52	0.03	99.95
8	39	0.02	99.97
9	21	0.01	99.98
10	10	0.01	99.99
11	13	0.01	99.99
12	3	0.00	99.99
13	6	0.00	100.00
15	1	0.00	100.00
17	1	0.00	100.00
18	1	0.00	100.00
26	1	0.00	100.00
Total	181,489	100.00	

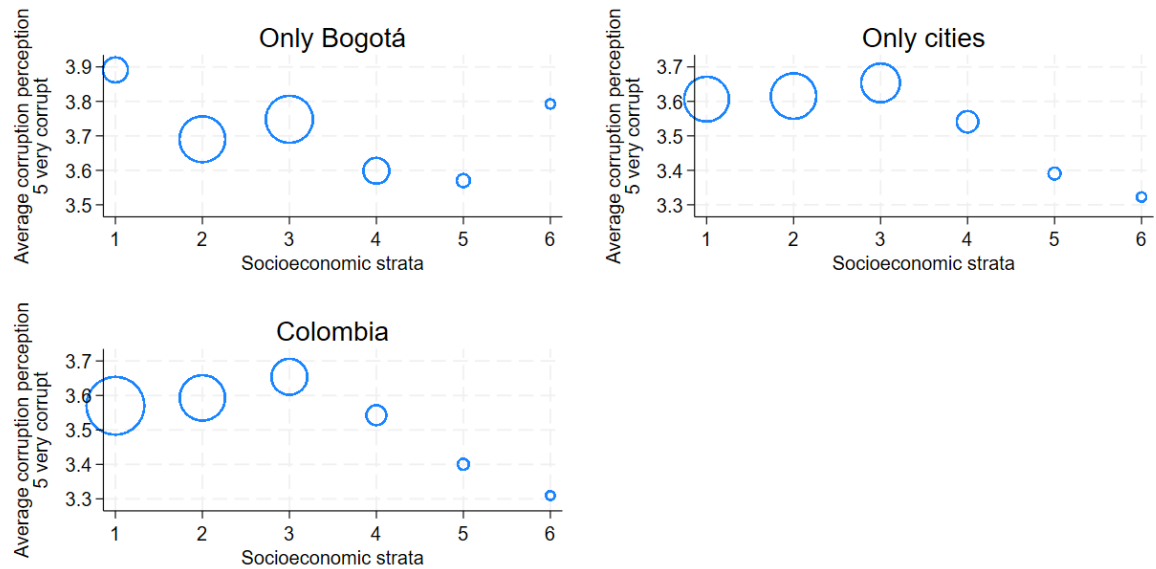
Table1. Times each plate passed exceeding the limit under the Cameras.

Perception of corruption of the Mayor's office By socioeconomic strata



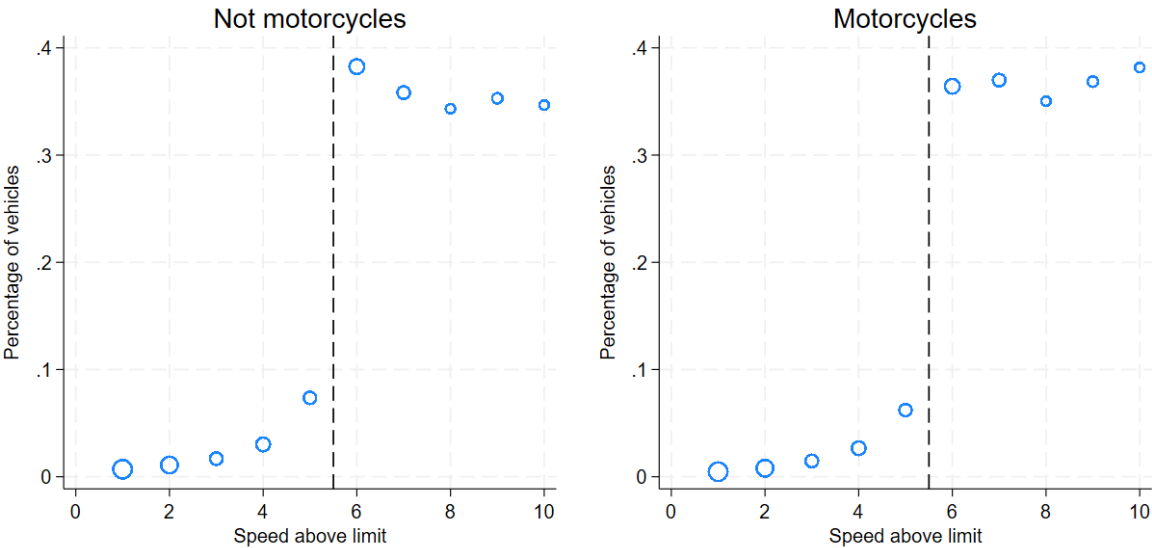
Question asks to rank from 1(Not corrupt at all) to 5 (Very corrupt)

Perception of corruption of the police By socioeconomic strata



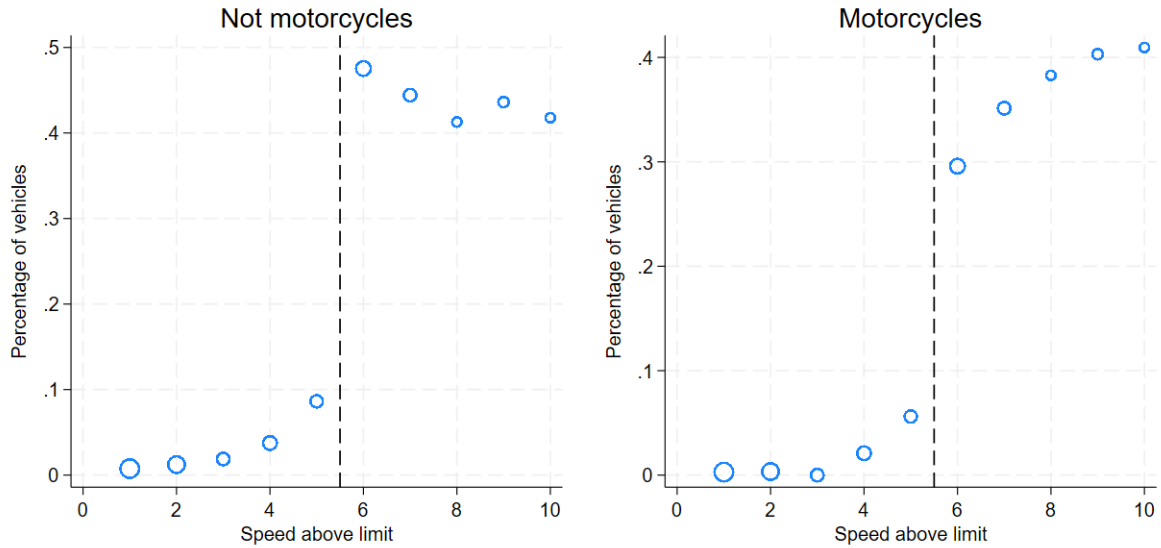
Question asks to rank from 1(Not corrupt at all) to 5 (Very corrupt)

Percentage of vehicles that receive a speeding ticket
by vehicle type
Full sample

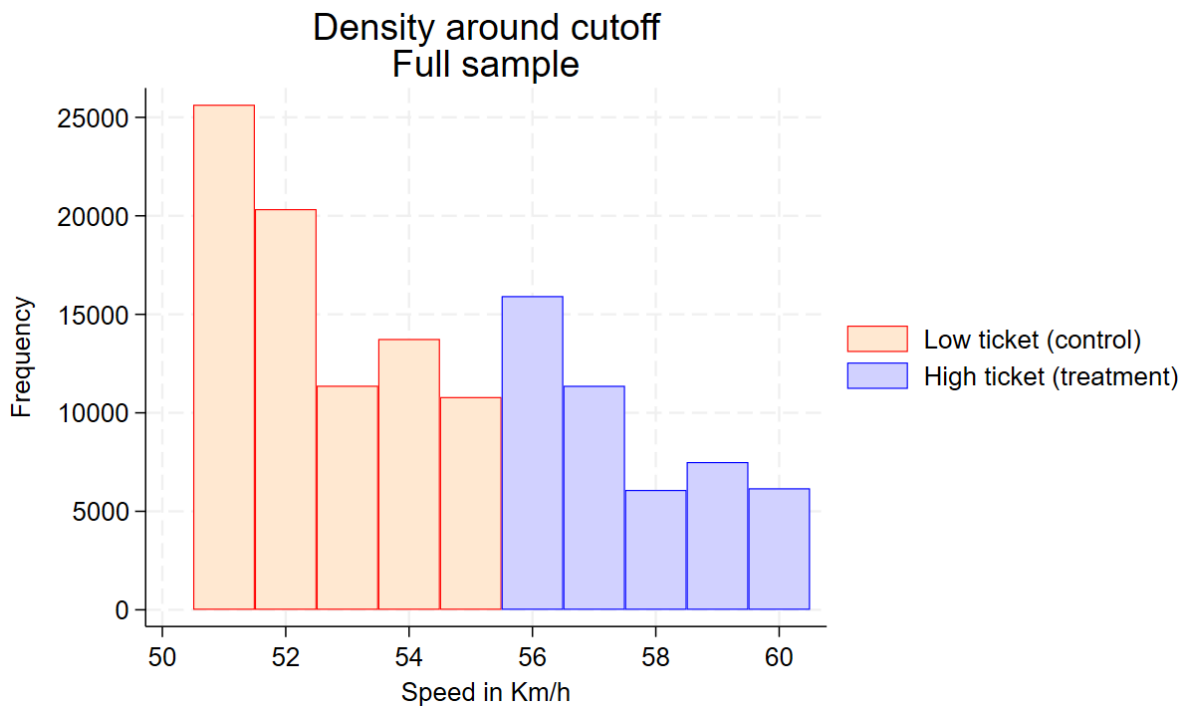


Graph 2.

Percentage of vehicles that receive a speeding ticket by vehicle type Matched sample



Graph 3.



2. Vehicles that passed more than once.

In my main estimation I use only vehicles in my sample that pass under a Camera once. This because observed speed compared to the real limit is the variable that randomly assigns vehicles to treatment, thus if a vehicle passes once below the real limit and once above the real limit is not clear to which condition of treatment it should be assigned. But this problem is not present when a vehicle passes always below or above the limit. In the following vehicles that fulfill this condition are also used and their assigned speed is the average of all the measures of speed. The date is the first time it passed above the legal limit under a Speed Camera.

This changes the Full sample for the estimation from 136533 to 144606, a 5% increase. For the matched sample it increases from 62288 to 76511, a 22% increase.

Results are essentially, and reassuringly, the same. In fact, they are slightly higher for the new Matched sample, particularly when it comes to the concentration of the effect in higher quintiles.

Full sample reoffending-speed:

	(1) Speeding	(2) TimesSpeed~g	(3) Speeding>10	(4) TimesSpee~10
Conventional	-0.140*** (-5.41)	-0.222*** (-3.68)	-0.00487 (-0.34)	-0.00261 (-0.11)
Bias-corrected	-0.167*** (-6.47)	-0.257*** (-4.25)	-0.0162 (-1.13)	-0.0145 (-0.64)
Robust	-0.167*** (-4.29)	-0.257*** (-2.82)	-0.0162 (-0.75)	-0.0145 (-0.43)
N	144606	144606	144606	144606
Mean	0.249	0.434	0.0636	0.0869

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 1 Appendix (equivalent to Table 8 of the Article). Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had a at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had a at least one serious speeding.

Matched sample reoffending-speed:

	(1) Speeding	(2) TimesSpeeding	(3) Speeding>10	(4) TimesSpeeding>10
Conventional	-0.132*** (-4.03)	-0.215*** (-3.07)	0.00648 (0.37)	0.00514 (0.20)
Bias-corrected	-0.151*** (-4.62)	-0.206*** (-2.94)	-0.0168 (-0.97)	-0.0180 (-0.72)
Robust	-0.151*** (-3.07)	-0.206* (-1.96)	-0.0168 (-0.65)	-0.0180 (-0.49)
N	65860	65860	65860	65860
Mean	0.274	0.444	0.0623	0.0771

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 2 Appendix (equivalent to Table 11 of the Article). Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

There is a slight difference with the main estimation, since in this case the effect is concentrated on quintile 5, instead of quintile 4. But anyways the main result holds, the deterrent effect of a speeding ticket is higher for the higher quintiles.

Matched Sample Quintiles

	(1) Quintile1	(2) Quintile2	(3) Quintile3	(4) Quintile4	(5) Quintile5
Conventional	-0.0727 (-0.77)	-0.138** (-2.04)	-0.159** (-1.99)	-0.127** (-2.08)	-0.133* (-1.96)
Bias-corrected	-0.0596 (-0.63)	-0.0783 (-1.15)	-0.201** (-2.52)	-0.169*** (-2.77)	-0.183*** (-2.70)
Robust	-0.0596 (-0.42)	-0.0783 (-0.77)	-0.201* (-1.66)	-0.169* (-1.84)	-0.183* (-1.76)
N	7458	11970	12642	16735	17055
Mean	0.189	0.261	0.272	0.293	0.304

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 3 Appendix (equivalent to Table 16). Reoffending by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise.